

BlockChain and Cryptocurrencies

Block chain and Crypto Currencies
Dr. Ahlad Kumar

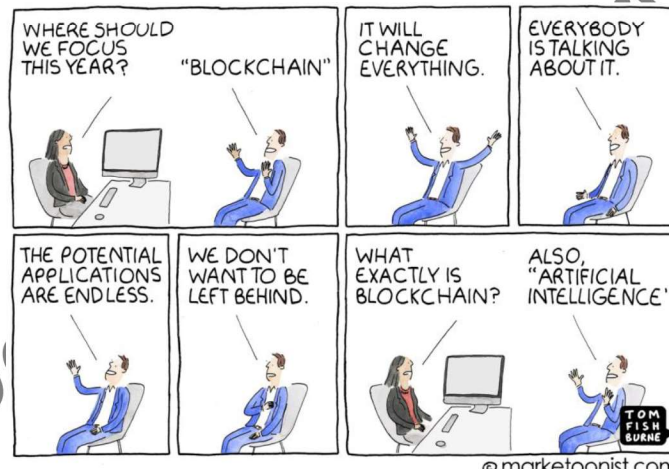
The Myth Busters

- Blockchain $\neq$ Bitcoin (or any other cryptocurrencies)
 - If you want to take this course to trade cryptocurrencies, this course is not for you !!
 - We do not want to make any comment on whether Bitcoin is good or whether Bitcoin should be blocked

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The Myth Busters

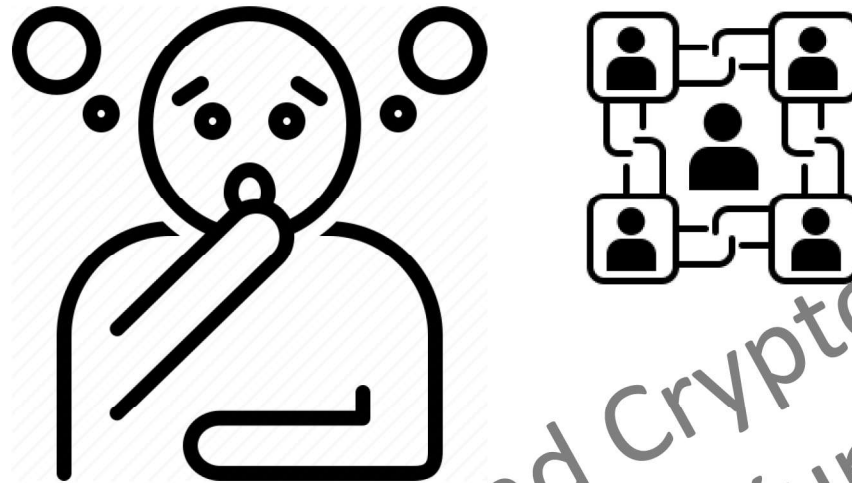
- Blockchain ≠ Bitcoin (or any other cryptocurrencies)
 - If you want to take this course to trade cryptocurrencies, this course is not for you !!
 - We do not want to make any comment on whether Bitcoin is good or whether Bitcoin should be blocked
- Anything and everything in the world cannot be solved using a blockchain
 - Blockchain is good but may not be so "stellar" the way it is projected



Then why should you take this course

- To avoid all the hypes and apply Blockchain as a solution at the right place ...

So, what is the right place?



Let's explore the course !!

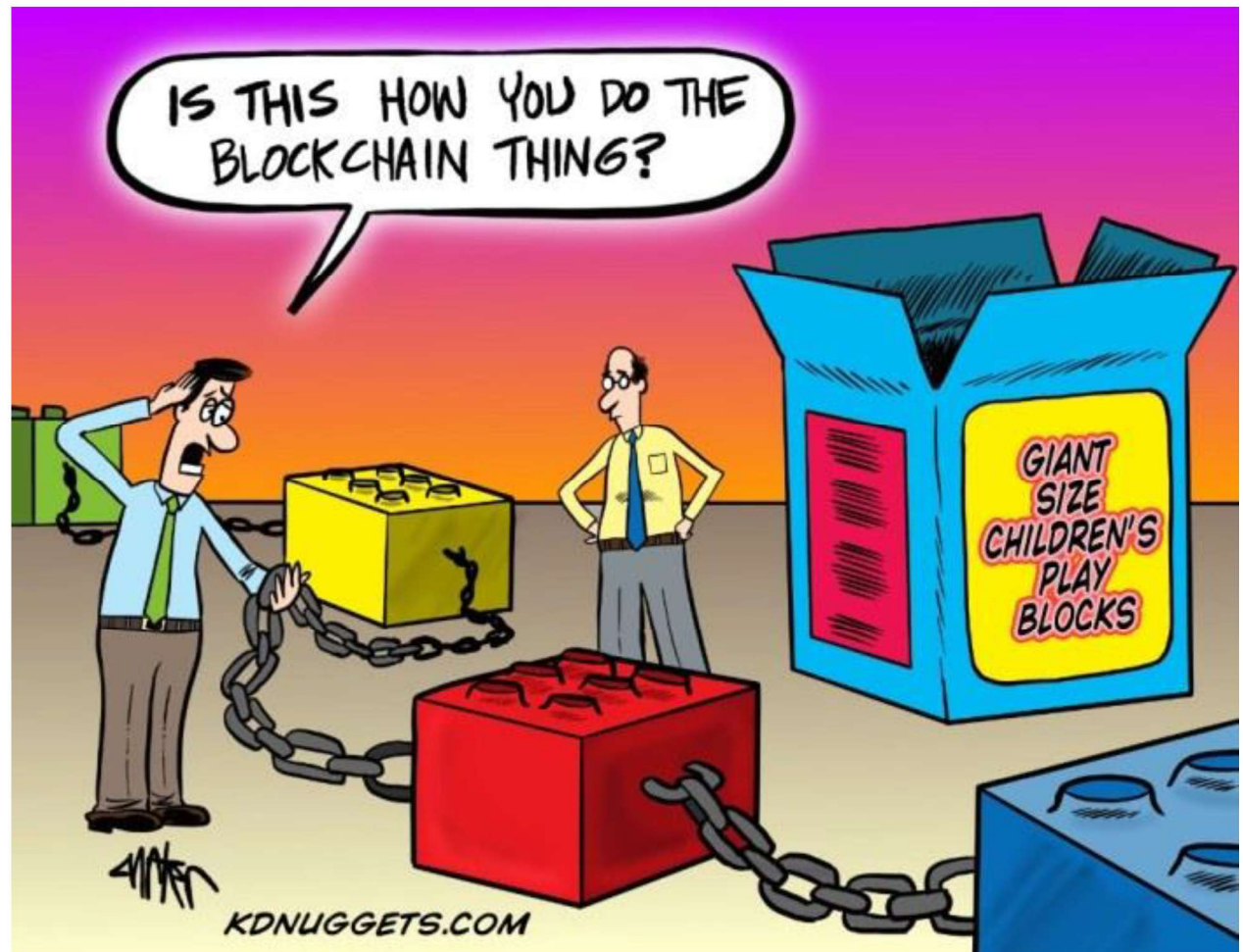
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What We'll Cover in This Course?

- Blockchain as a Data Structure
 - How does a blockchain look like?
 - How do we efficiently store data in a blockchain?
 - How can we efficiently manage data insertion in a blockchain? What is the complexity of data insertion and searching a data item within a blockchain?

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So, What is a
Blockchain?



Supply Chain Management: The Players and the Game

Supply Chain in Petroleum Industry



Crude Purchase

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Supply Chain in Petroleum Industry



Crude Purchase



Crude Transportation

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Supply Chain in Petroleum Industry



Crude Purchase



Crude Transportation



Crude Storage

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Crude Purchase



Crude Transportation



Crude Storage



Refining

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Supply Chain in Petroleum Industry



Crude Purchase



Crude Transportation



Crude Storage



Refining



Distribution

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Crypto Currents

Supply Chain in Petroleum Industry



Crude Purchase



Crude Transportation



Crude Storage



Refining



Distribution



Retail

Block chain
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Crypto Currency

Petroleum Supply Chain in India

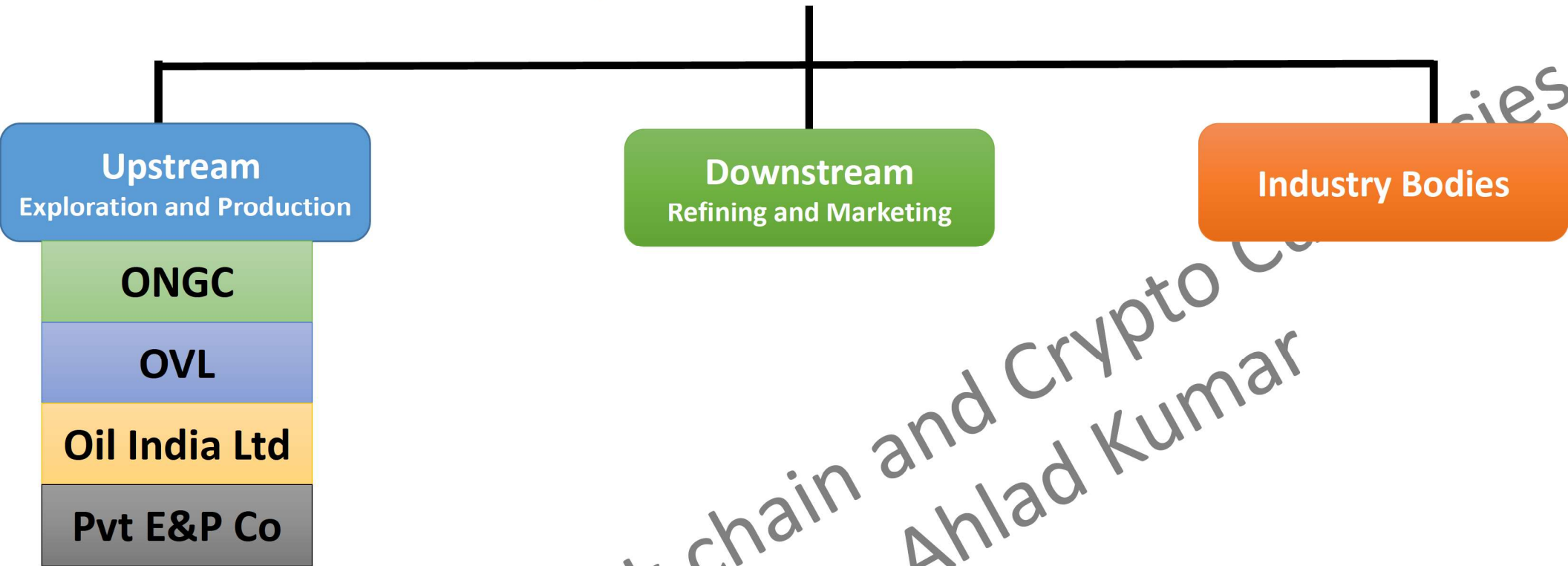
Ministry of Petroleum and Natural Gas



Diagram Source: <https://www.slideshare.net/naseer9848/supply-chain-and-logistics-issues-of-crude-oil-in-india>

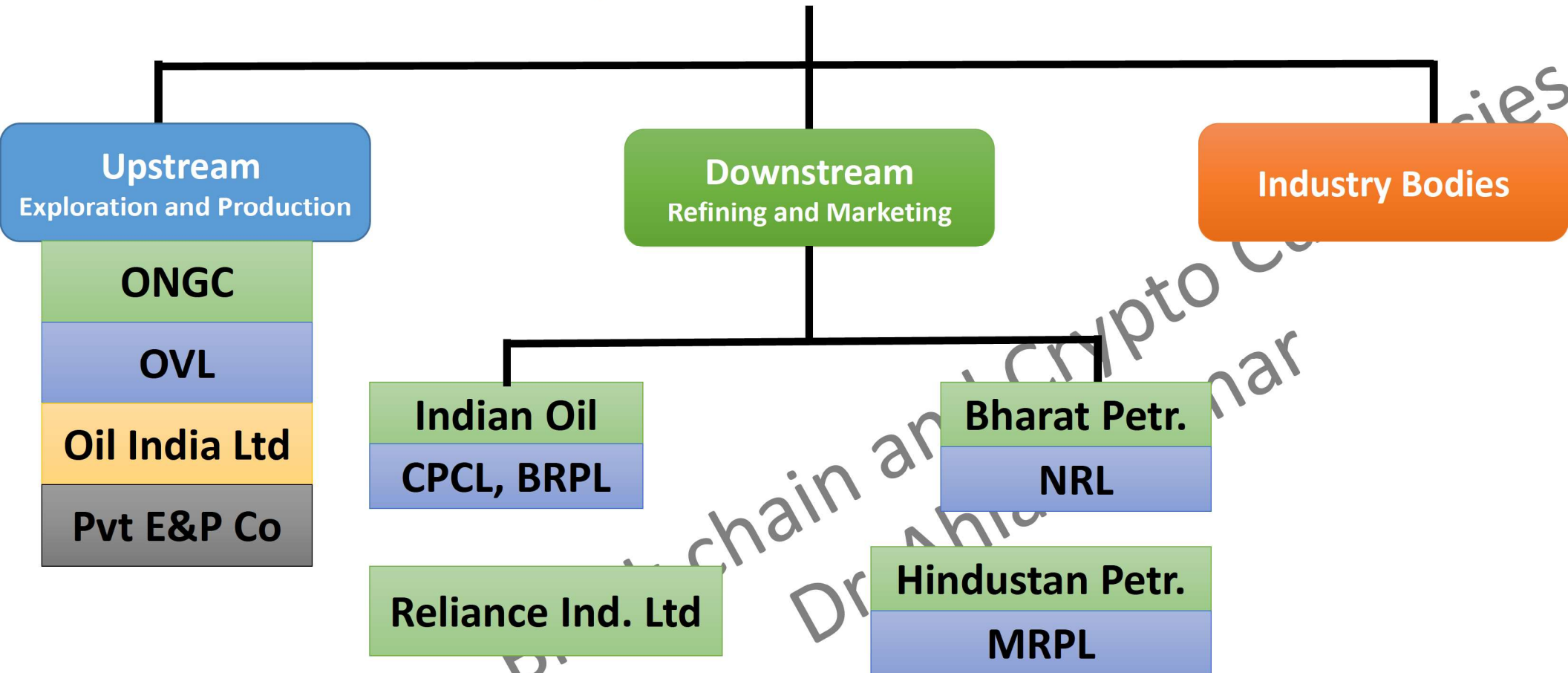
Petroleum Supply Chain in India

Ministry of Petroleum and Natural Gas



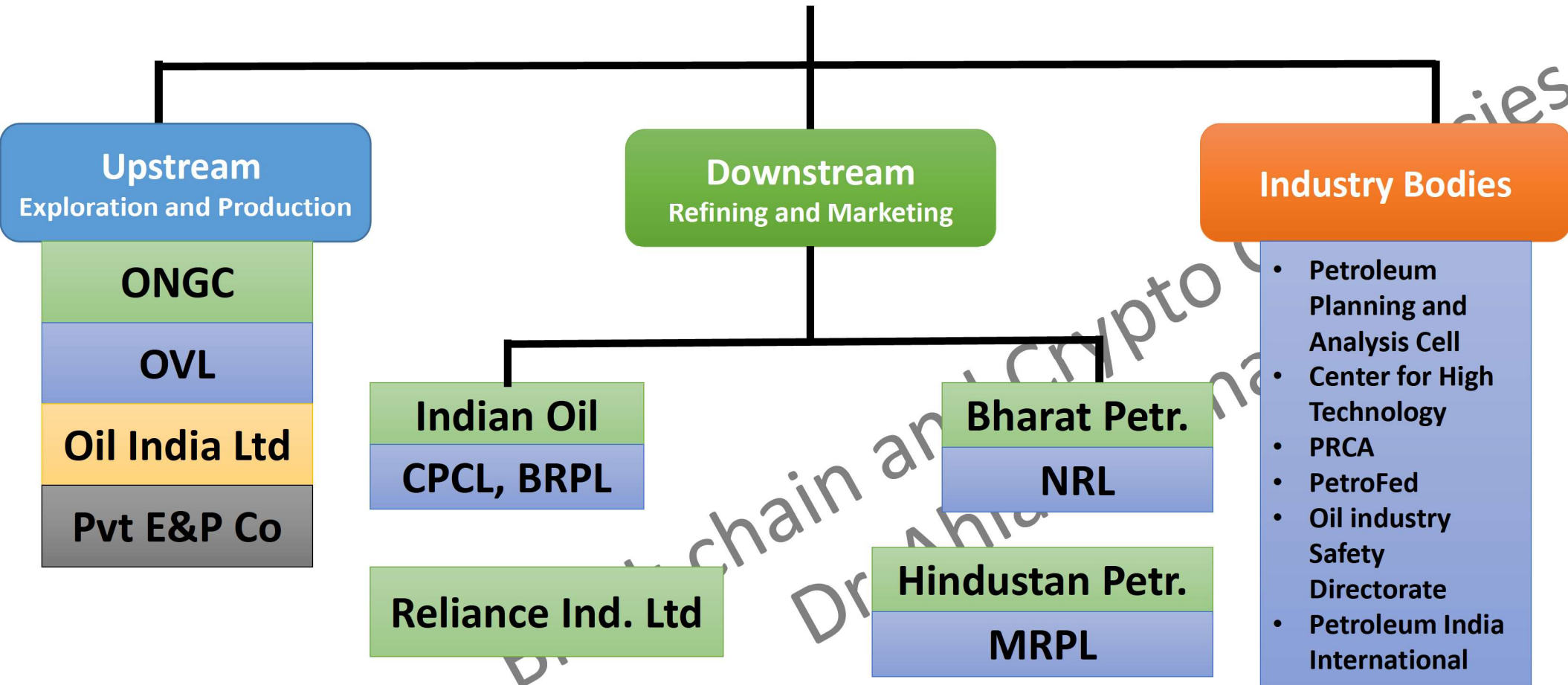
Petroleum Supply Chain in India

Ministry of Petroleum and Natural Gas



Petroleum Supply Chain in India

Ministry of Petroleum and Natural Gas



Requirements of a Successful Supply Chain

- Minimization of material procurement
- Maximization of manufacturing capacity and sales
- Meet demand numbers
- Respond quickly to market opportunity by purchasing the production shortfall from other players
- Objective of each production unit would be to maximize the throughput and its margin
- Procurement would purchase the feedstock with not the best yields at lowest cost

Information Source: <https://www.slideshare.net/naseer9848/supply-chain-and-logistics-issues-of-crude-oil-in-india>

Requirements of a Successful Supply Chain

- Minimization of material procurement



Needs Strong Coordination among the Players

its margin

- Procurement would purchase the feedstock with not the best yields at lowest cost

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Requirements of a Successful Supply Chain

- Minimization of material procurement



How do we obtain Real-time Information from the Stakeholders?

its margin

- Procurement would purchase the feedstock with not the best yields at lowest cost

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Requirements of a Successful Supply Chain

- Minimization of material procurement



How do we obtain Real-time Information from the Stakeholders?
A web-based portal?

Information Source: <https://www.slideshare.net/naseer9848/supply-chain-and-logistics-issues-of-crude-oil-in-india>

Requirements of a Successful Supply Chain

- Minimization of material procurement



How do we obtain Real-time Information from the Stakeholders?
What is the guarantee that the information submitted is correct?

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How do we obtain Real-time Information from the Stakeholders?
What is the guarantee that the information submitted is correct?
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How do we obtain Real-time Information from the Stakeholders?

What is the guarantee that the information submitted is correct?

What if someone denies the information later on?

We need a decentralized solution where no one trust each other but they should co-operate

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Requirements of a Successful Supply Chain

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How do we obtain Real-time Information from the Stakeholders?

What is the guarantee that the information submitted is correct?

What if someone denies the information later on?

We need a decentralized solution where no one trusts each other but they should cooperate

Blockchain is the answer !!

Conclusion

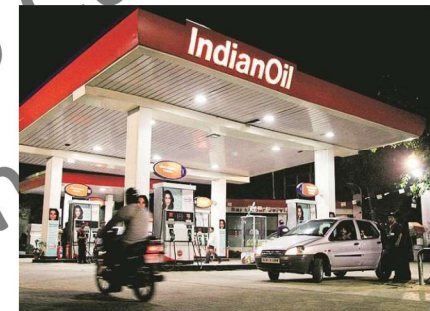
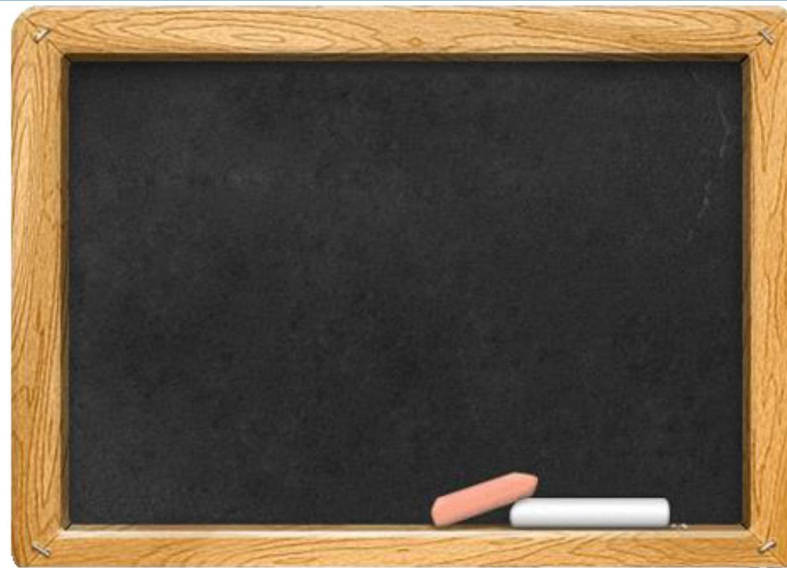
- You have a network of different players) businesses, enterprises, commercial establishments, government or private bodies or even individuals)
- Everyone has their own interest they want to fulfill their goal
- They do not trust each other
- If they cooperate, the society will get benefited
- Trustless Decentralization: Blockchain

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How Can We Obtain Real Time Information?



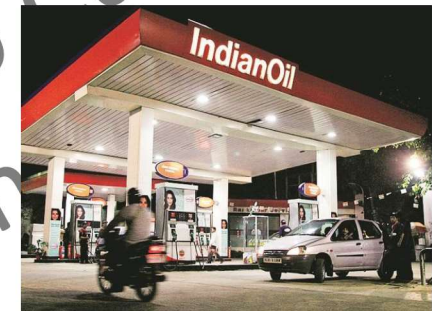
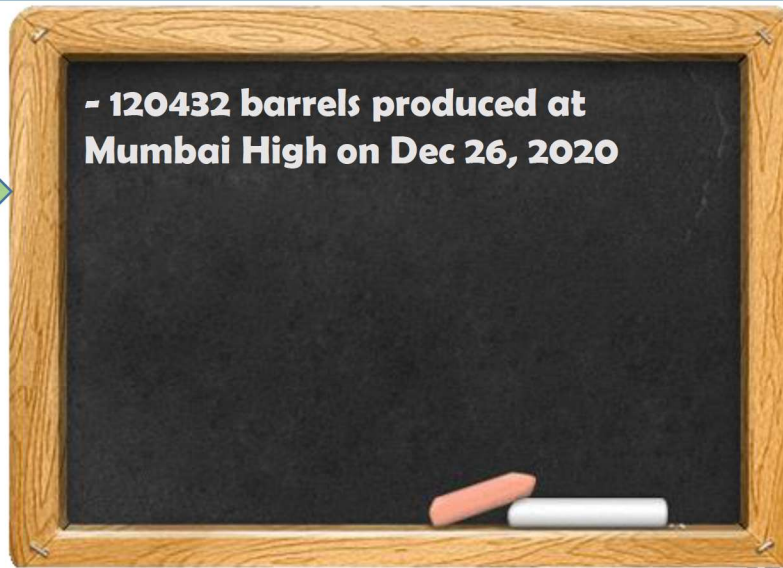
Use a Public Bulletin Board



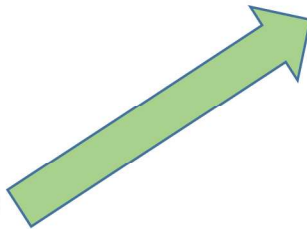
Use a Public Bulletin Board



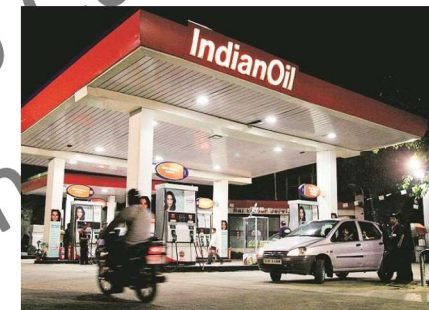
**- 120432 barrels produced at
Mumbai High on Dec 26, 2020**



Use a Public Bulletin Board



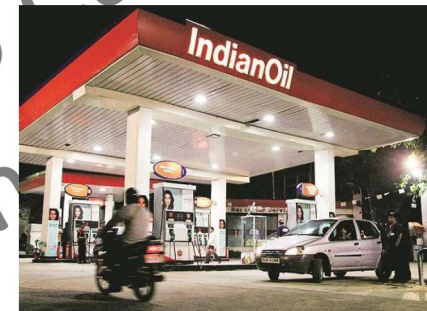
**- 120432 barrels produced at
Mumbai High on Dec 26, 2020**
**- 16467 barrels transported from
Mumbai High to HPCL Refinery
on Dec 26, 2020 at 2:30 pm**



Use a Public Bulletin Board



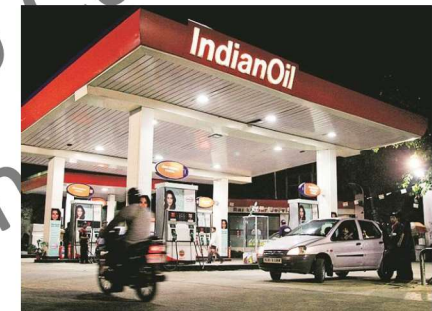
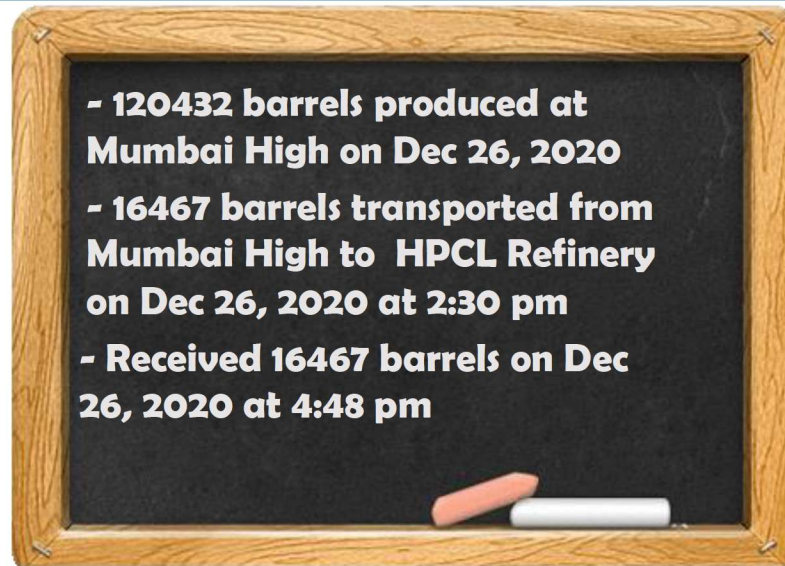
- 120432 barrels produced at Mumbai High on Dec 26, 2020
- 16467 barrels transported from Mumbai High to HPCL Refinery on Dec 26, 2020 at 2:30 pm
- Received 16467 barrels on Dec 26, 2020 at 4:48 pm



Use a Public Bulletin Board -- Advantages



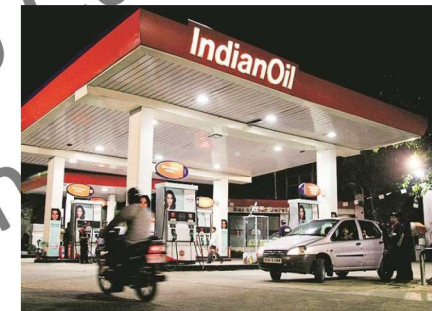
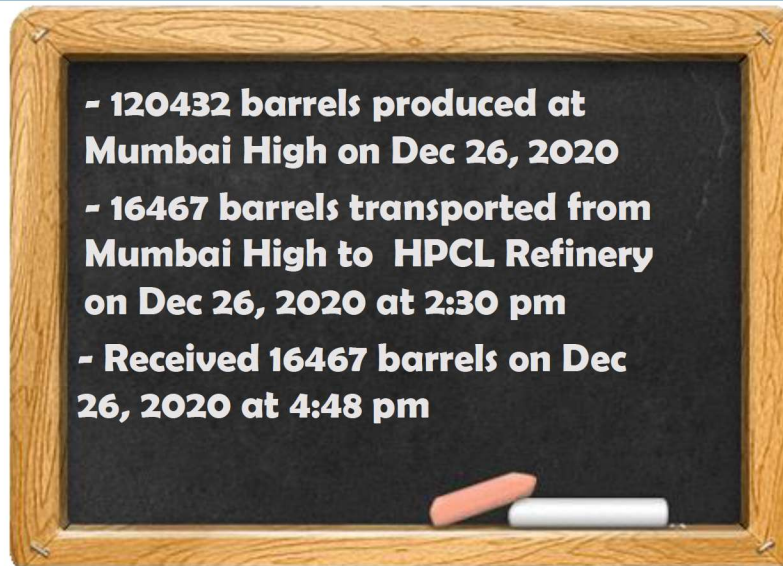
- Everyone can see all the logs and verify
- Infinite space to write



Use a Public Bulletin Board -- Advantages



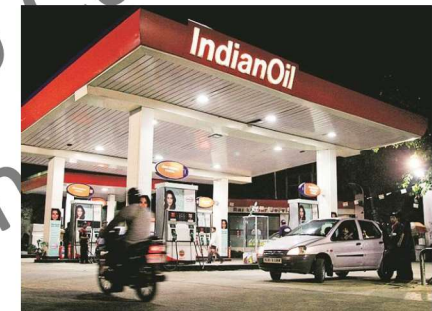
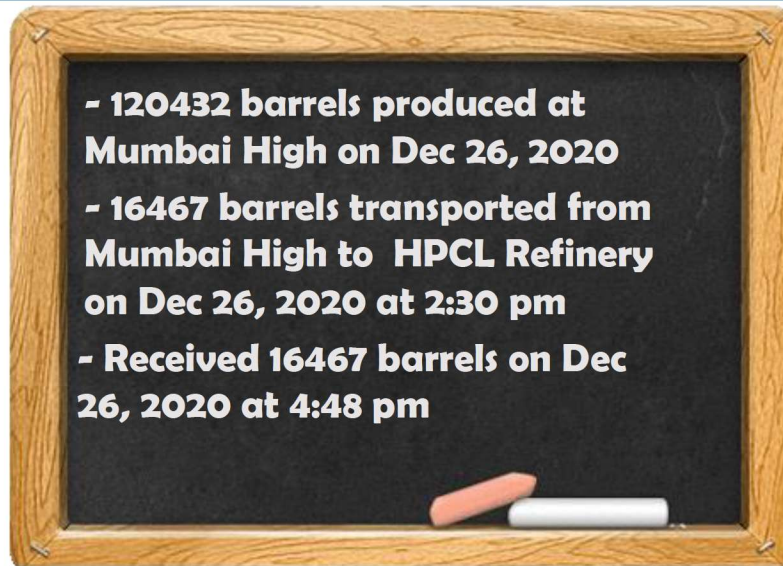
- Any change in information is visible to everyone



Use a Public Bulletin Board -- Advantages



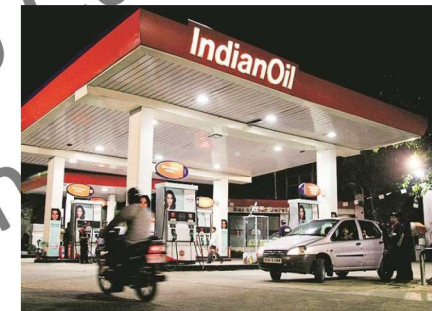
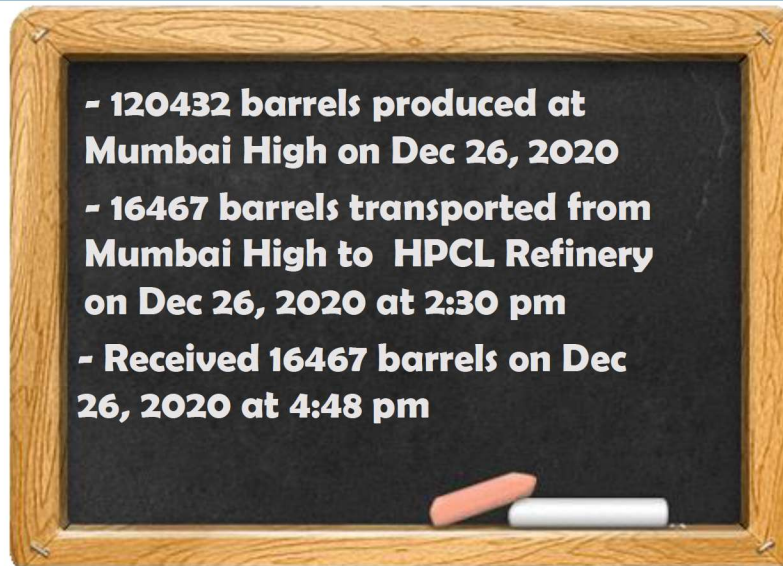
- The board is not erasable, no one can deny later



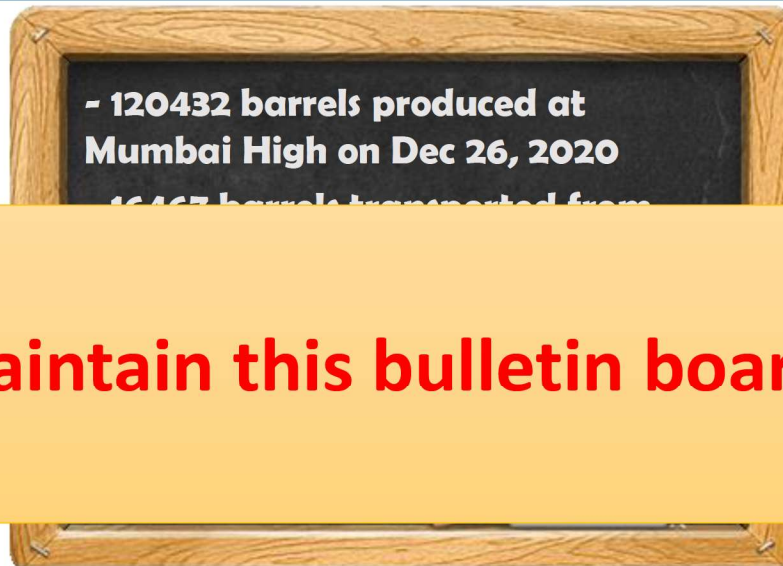
Use a Public Bulletin Board -- Advantages



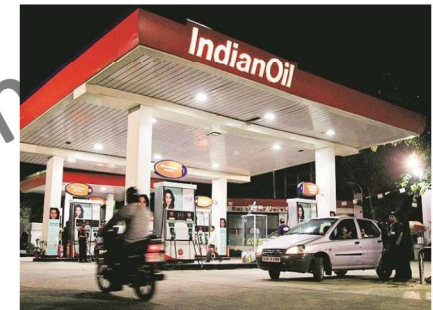
- **Simple one-step auditing**



Use a Public Bulletin Board -- Challenges



Who will maintain this bulletin board?



Blockchain and Cryptocurrency

Use a Public Bulletin Board -- Challenges



Who will maintain this bulletin board?

- Buy Cloud from amazon



Blockchain

Use a Public Bulletin Board -- Challenges



Who will maintain this bulletin board?

- Buy Cloud from amazon

Who will manage it and provide the cost?



Blockchain

Use a Public Bulletin Board -- Challenges



Who will maintain this bulletin board?

- Buy Cloud from amazon
- One of the enterprises maintain a private cloud



Blockchain

Use a Public Bulletin Board -- Challenges



Who will maintain this bulletin board?

- Buy Cloud from amazon
 - One of the enterprises maintain a private cloud
- What is the guarantee that it is not a fraud?



Blockchain

Use a Public Bulletin Board -- Challenges



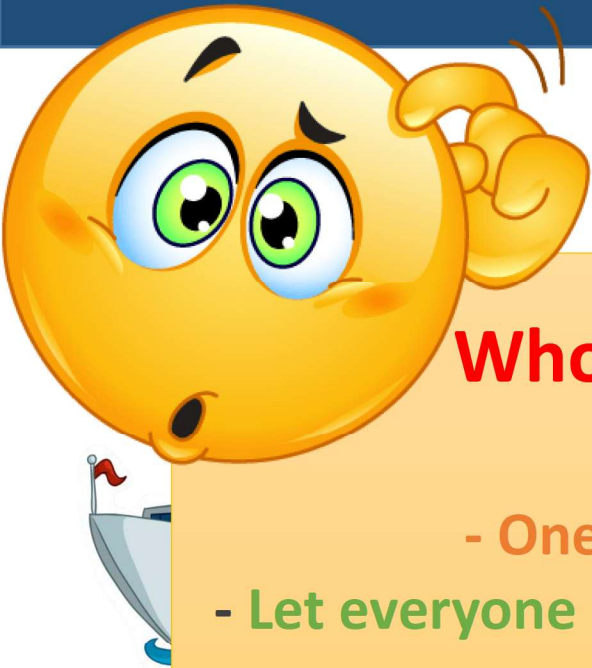
Who will maintain this bulletin board?

- Buy Cloud from amazon
- One of the enterprises maintain a private cloud
- Let everyone maintain the same copy of the board individually and independently



Blockchain

Use a Public Bulletin Board -- Challenges



Who will maintain this bulletin board?

- Buy Cloud from amazon
- One of the enterprises maintain a private cloud
- Let everyone maintain the same copy of the board individually and independently – **BUT HOW?**



Blockchain

Use a Public Bulletin Board -- Challenges



Who will maintain this bulletin board?

- Buy Cloud from amazon
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Blockchain

What is this “Blockchain”?



**A decentralized and
multi-authority
networked information
data storage and access
system**

and Crypto Currencies
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What is this “Blockchain”?

- 120432 barrels produced at
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No one is the sole-owner of the data, but everyone has a copy of the data - there is no central database

What is this “Blockchain”?



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Everyone holds
exactly the same
copy of the data at
the same instance
of the time

What is this “Blockchain”?



**An immutable
append-only ever-
growing chain of
data. Data once
added cannot be
deleted or
modified later**

What is this “Blockchain”?



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What is this “Blockchain”?



**An immutable
append-only ever-
growing chain of data.
Data once added
cannot be deleted or
modified later**



**Once something is
added in the
blockchain, it cannot
be denied later**

What is this “Blockchain”?



The information is transparent to all - everyone can see what is going on in the system



What is this “Blockchain”?



The information is transparent to all - everyone can see what is going on in the system



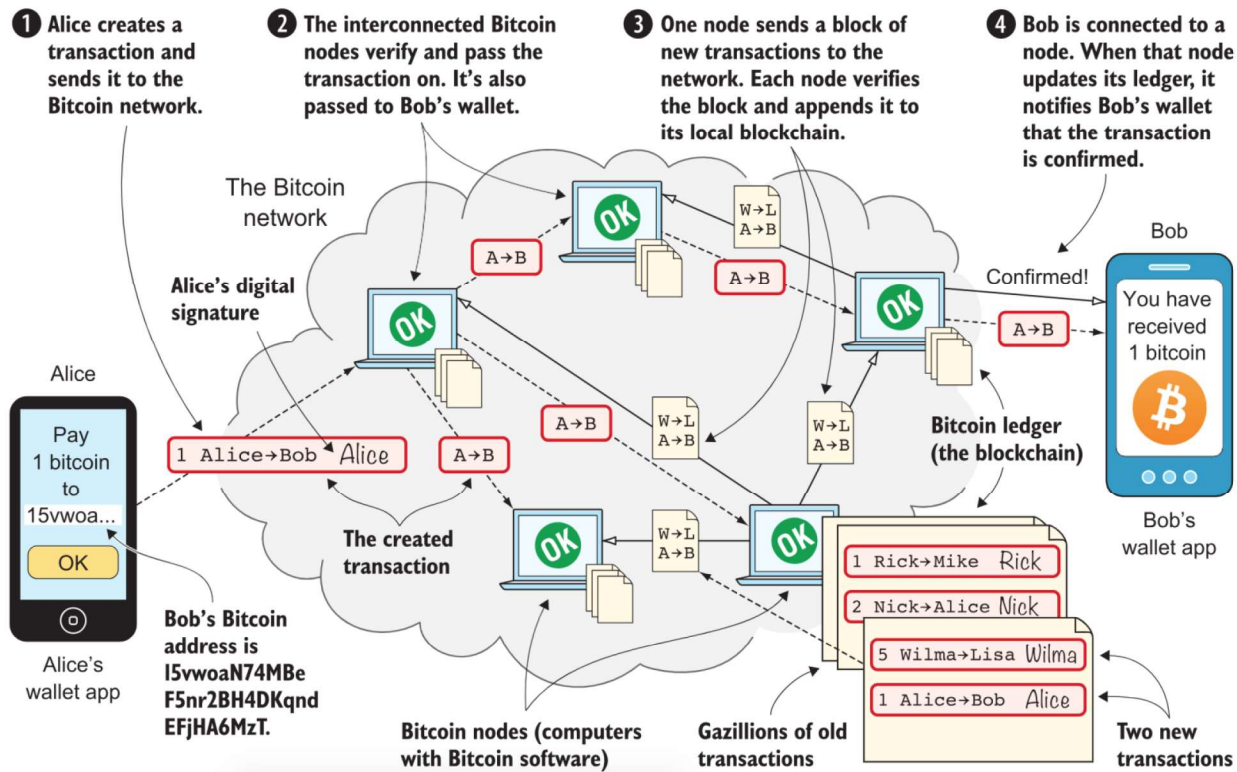
No-one can make any change without others to notice it



So, What Can be the Definition of a "Blockchain"

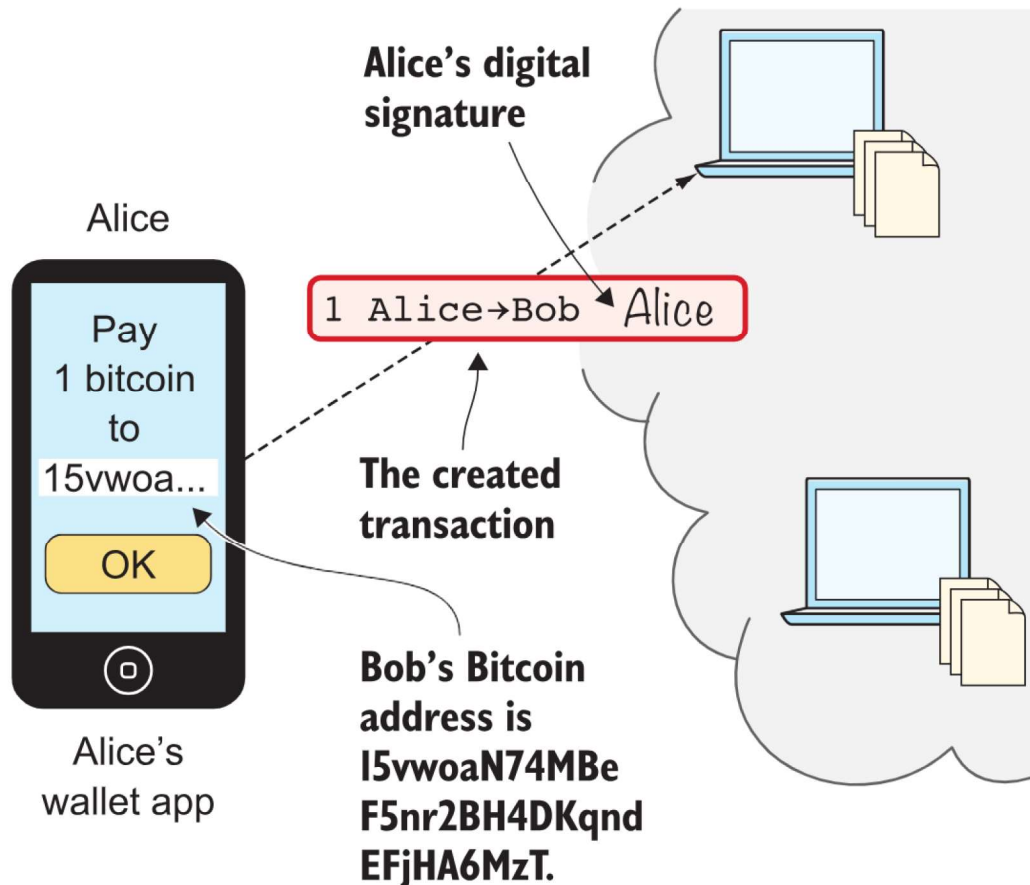
**A decentralized immutable append-only
public ledger**

Big Picture



- (1) Alice creates and signs a transaction that moves 1 bitcoin from her to Bob. She then sends the transaction to the Bitcoin network.
- (2) The computers in the network check that Alice actually has the money to spend and that the transaction is authentic. They then pass the transaction to their neighbours, called *peers*.
- (3) Each computer updates its own copy of the *Bitcoin blockchain*, or the *ledger*, with the new payment information.
- (4) The network notifies Bob that he has received 1 bitcoin.

(1) Transactions



The Bitcoin *transaction* is a piece of data specifying

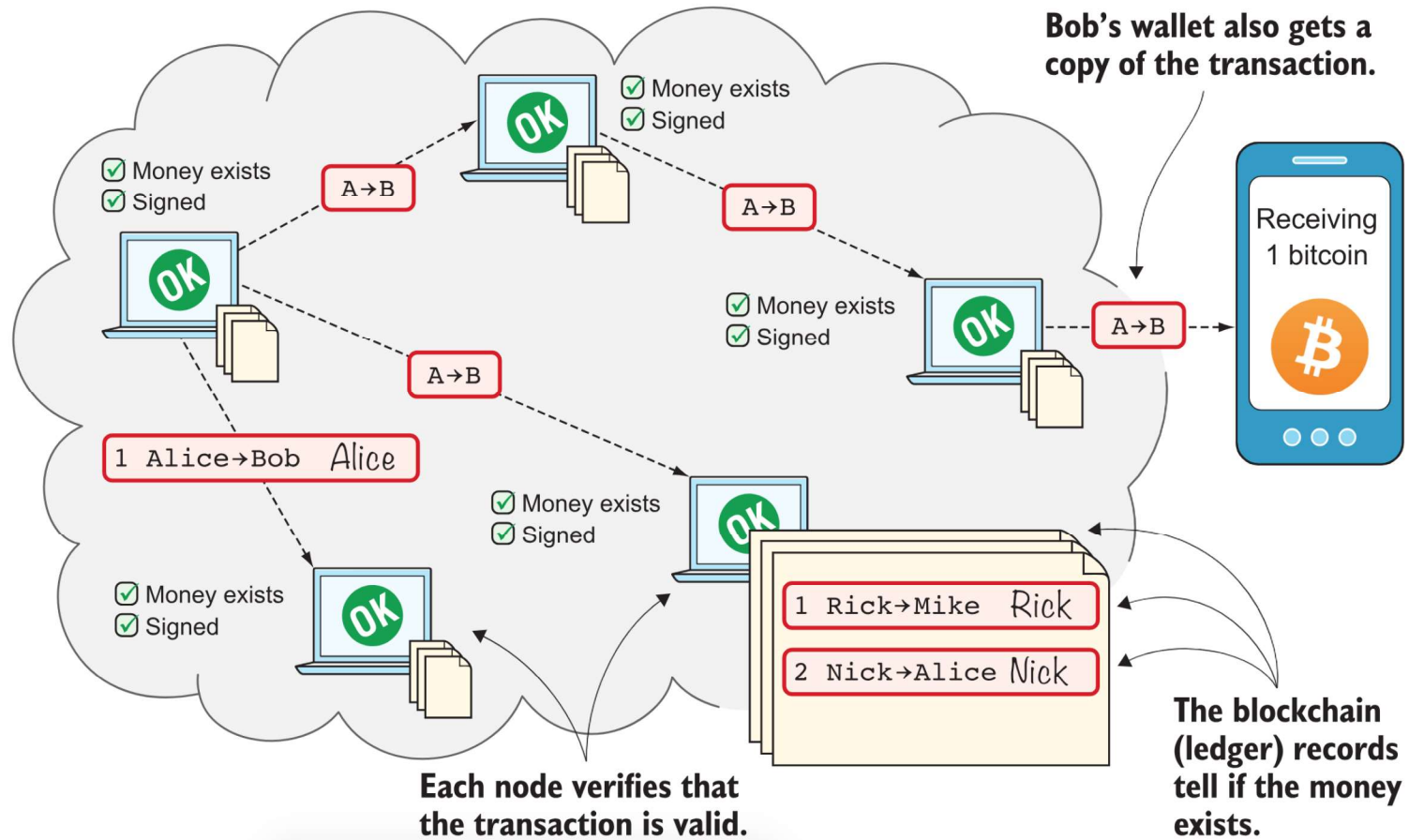
(1) The amount to move (1 bitcoin)

(2) The Bitcoin address to move the money to (Bob's Bitcoin address)

15vwoaN74MBeF5nr2BH4DKqndEFjHA6MzT

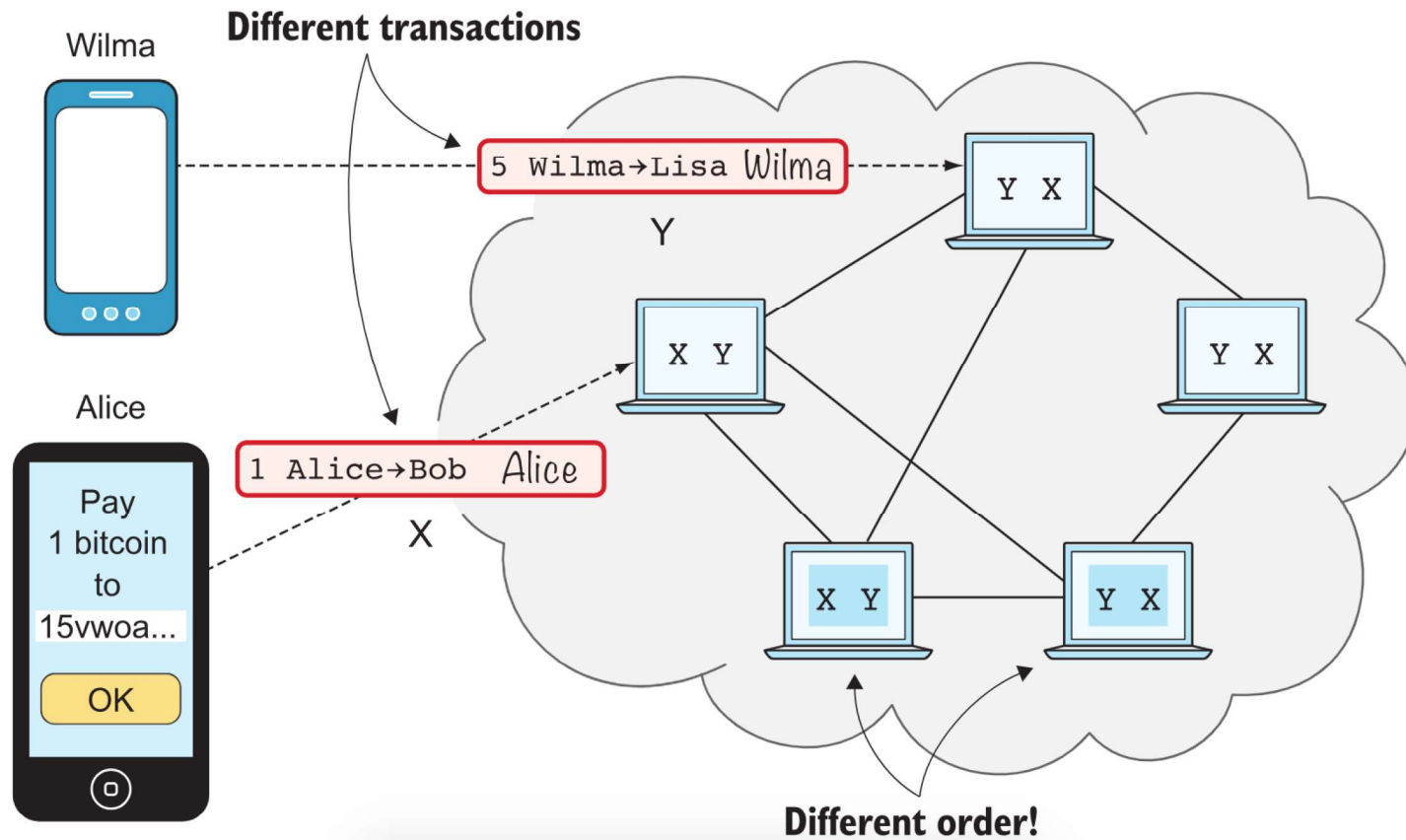
(3) A *digital signature* (made with Alice's private key)

The Blockchain network



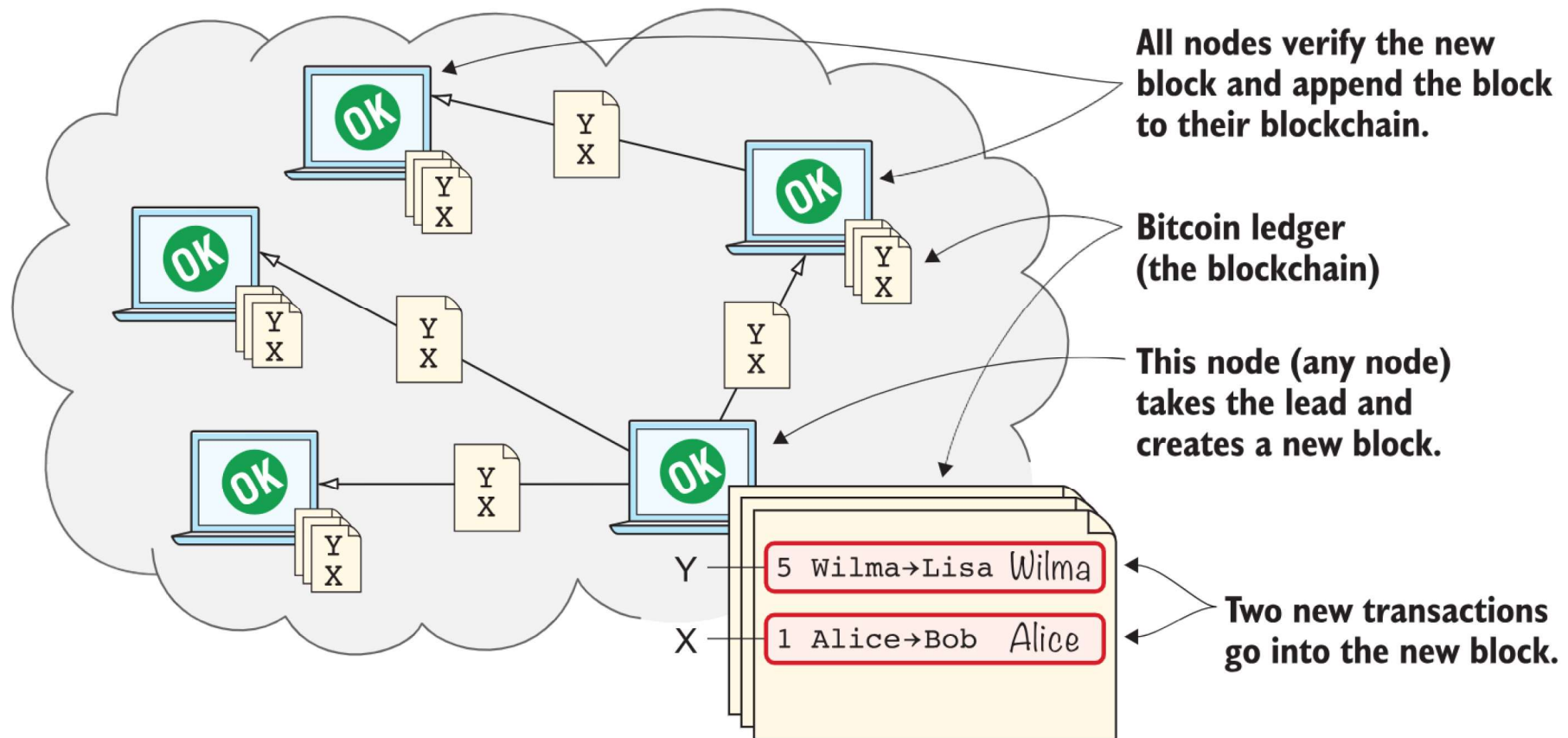
Alice has sent her transaction to a node in the network. The node verifies the transaction and forwards it to other nodes. Eventually, the transaction will reach all nodes in the network.

The Blockchain



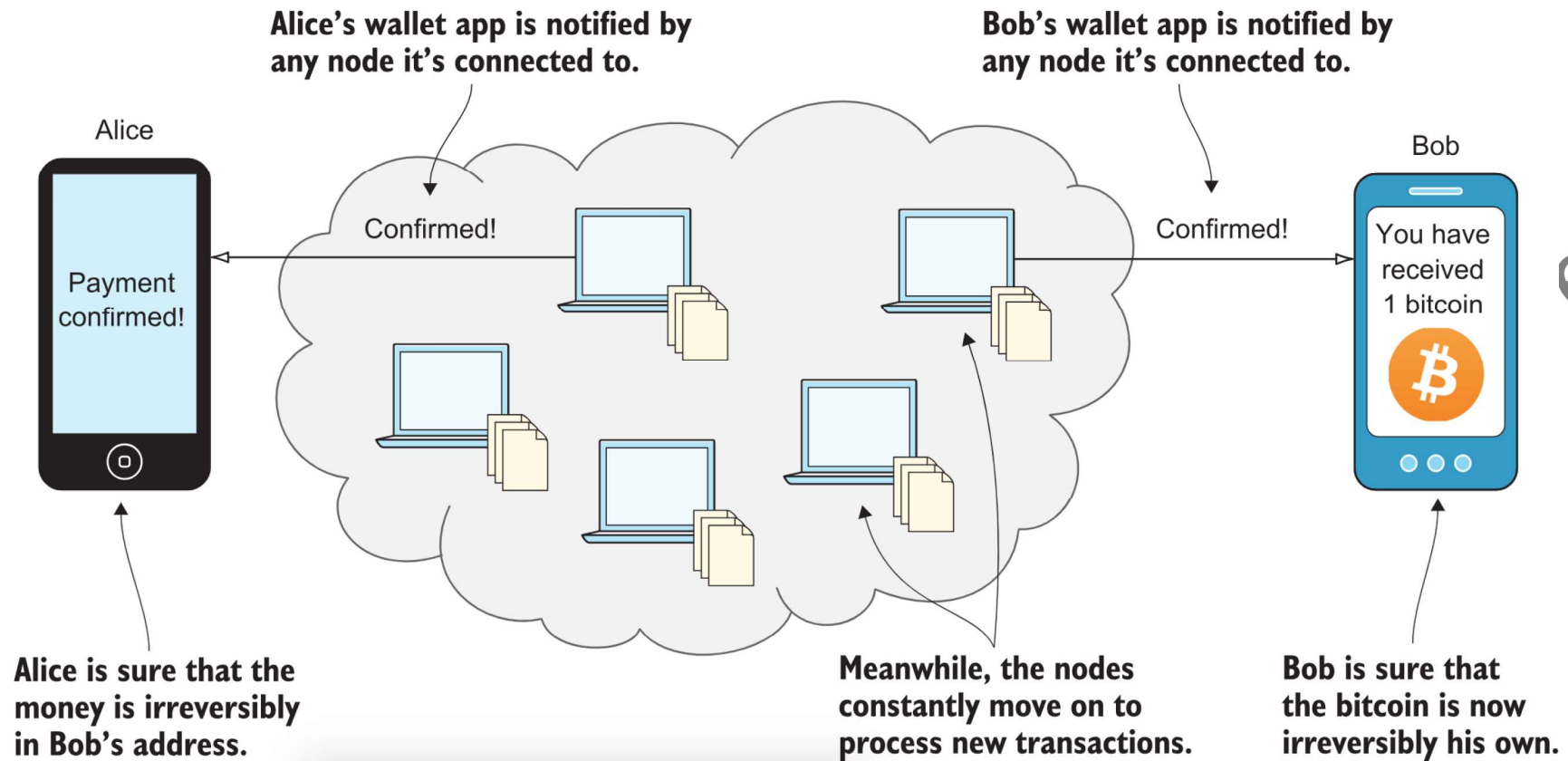
Transactions arrive in different orders at different nodes. If all nodes wrote their transactions to the blockchain in order of arrival, the different nodes blockchains would differ.

The Blockchain network



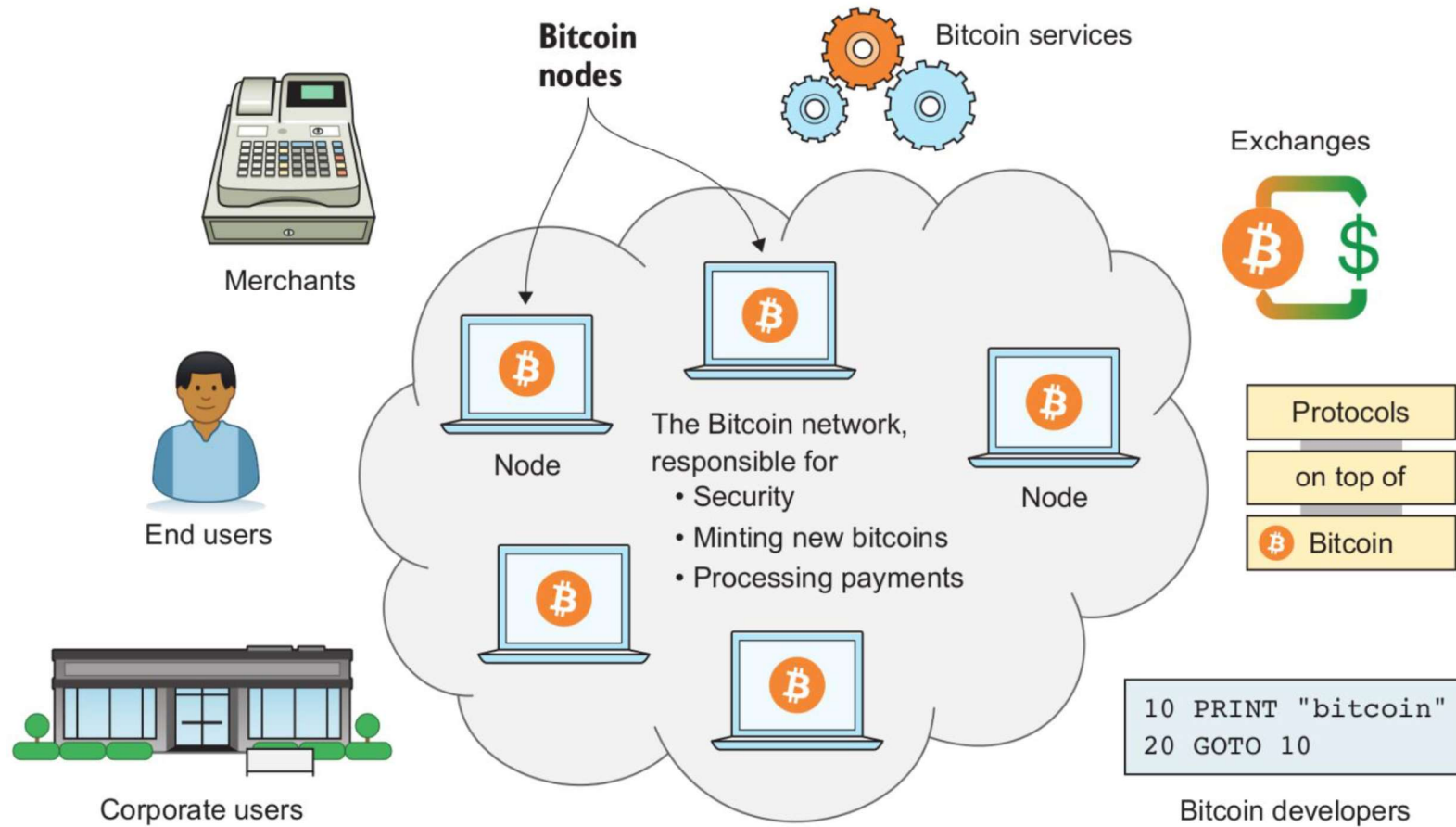
One node takes the lead and tells the others in what order to add transactions. The other nodes verify the block and update their blockchain copies accordingly.

Wallets



Bob's wallet has asked a node to notify the wallet upon activity at Bob's Bitcoin address. Alice pays to Bob's address, and the node has just written the transaction to the blockchain, so it notifies Bob's wallet.

Customers



- **End users:** People using Bitcoin for their day-to-day needs, such as savings, shopping, speculation, or salaries
- **Corporate users :** Companies using Bitcoin to solve their business needs, such as paying wages internationally, or use cases similar to those of end users
- **Merchants:** For example, a restaurant or a bookstore accepting Bitcoin payments
- **Bitcoin services:** Companies providing Bitcoin-related services to customers, such as topping up mobile phones, anonymization services, remittance services, or tipping services
- **Exchanges :** Commercial services people can use to exchange their local currency to and from bitcoins
- **Protocols on top:** Systems that operate “on top” of Bitcoin to perform certain tasks, such as payment network protocols, specialized tokens, and decentralized exchanges
- **Bitcoin developers:** People working, often for free, with the open source computer programs that participants of the Bitcoin network use